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Experiment - 1

Experiment Name

Simple Linear Regression — Salary Prediction

Question Statement

Build a Simple Linear Regression model to predict Salary based on Years of Experience.

Input / Dataset

Dataset: Salary_Data.csv

Python Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression
```

Output

R^2 Score : 0.9745

Graph / Visualization

Regression graph showing salary vs experience

Experiment - 2

Experiment Name

Linear Regression — House Price Prediction

Question Statement

Create a Linear Regression model for House Price Prediction.

Input / Dataset

Dataset: House_Price.csv

Python Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression
```

Output

R^2 Score : 0.9512

Graph / Visualization

Regression line and residual plot

Experiment - 3

Experiment Name

Hypothesis Testing in Linear Regression

Question Statement

Perform hypothesis testing in Linear Regression.

Input / Dataset

Dataset: Student_Marks.csv

Python Code

```
import statsmodels.api as sm
```

Output

```
p-value : 0.000001
```

Graph / Visualization

Study hours vs marks graph

Experiment - 4

Experiment Name

Multiple Linear Regression

Question Statement

Implement Multiple Linear Regression using sklearn.

Input / Dataset

Dataset: House_MLR.csv

Python Code

```
from sklearn.linear_model import LinearRegression
```

Output

R^2 Score : 0.9623

Graph / Visualization

Residual analysis visualization

Experiment - 5

Experiment Name

Feature Selection using RFE

Question Statement

Perform Feature Selection using Recursive Feature Elimination.

Input / Dataset

Dataset: Housing_Features.csv

Python Code

```
from sklearn.feature_selection import RFE
```

Output

Selected Features: Area, Rooms, Locality_Score

Graph / Visualization

Feature ranking graph

Experiment - 6

Experiment Name

Linear Regression with Categorical Variables

Question Statement

Build Linear Regression using Dummy Encoding.

Input / Dataset

Dataset: House_Location.csv

Python Code

```
pd.get_dummies(df)
```

Output

R^2 improved to 0.9541

Graph / Visualization

Category-wise comparison graph

Experiment - 7

Experiment Name

Detecting Multicollinearity using VIF

Question Statement

Analyze multicollinearity using VIF.

Input / Dataset

Dataset: House_VIF.csv

Python Code

```
variance_inflation_factor()
```

Output

VIF reduced below 2

Graph / Visualization

Correlation heatmap

Experiment - 8

Experiment Name

Logistic Regression — Customer Churn Prediction

Question Statement

Predict customer churn using Logistic Regression.

Input / Dataset

Dataset: Telecom_Churn.csv

Python Code

```
from sklearn.linear_model import LogisticRegression
```

Output

```
Accuracy : 0.8500
```

Graph / Visualization

Confusion matrix graph

Experiment - 9

Experiment Name

Logistic Regression — Evaluation Metrics

Question Statement

Evaluate classification metrics.

Input / Dataset

Dataset: Binary_Classification.csv

Python Code

```
confusion_matrix()
```

Output

F1-Score : 0.8485

Graph / Visualization

Metrics comparison graph

Experiment - 10

Experiment Name

ROC Curve and Optimal Threshold

Question Statement

Plot ROC curve and determine threshold.

Input / Dataset

Dataset: Binary_Classification.csv

Python Code

```
roc_curve()
```

Output

```
AUC Score : 0.9127
```

Graph / Visualization

ROC curve visualization

Experiment - 11

Experiment Name

Naive Bayes Classification — Manual Calculation

Question Statement

Implement Naive Bayes manually.

Input / Dataset

Custom Email Dataset

Python Code

```
Bayes theorem calculations
```

Output

Classification : SPAM

Graph / Visualization

Posterior probability graph

Experiment - 12

Experiment Name

SMS Spam Ham Classifier

Question Statement

Build Spam/Ham classifier using Multinomial NB.

Input / Dataset

Custom SMS Dataset

Python Code

```
MultinomialNB()
```

Output

Accuracy : 1.0000

Graph / Visualization

Spam indicator visualization

Experiment - 13

Experiment Name

Bernoulli Naive Bayes Comparison

Question Statement

Compare BernoulliNB and MultinomialNB.

Input / Dataset

Custom SMS Dataset

Python Code

```
BernoulliNB()
```

Output

Accuracy : 0.9333

Graph / Visualization

Model comparison graph

Experiment - 14

Experiment Name

K-Means Clustering — Customer Segmentation

Question Statement

Perform customer segmentation using K-Means.

Input / Dataset

Dataset: Customer_Segmentation.csv

Python Code

```
from sklearn.cluster import KMeans
```

Output

3 clusters generated

Graph / Visualization

Elbow method graph

Experiment - 15

Experiment Name

Hierarchical Clustering — Dendrogram Analysis

Question Statement

Perform Hierarchical Clustering.

Input / Dataset

Custom 2D Dataset

Python Code

```
from scipy.cluster.hierarchy import dendrogram
```

Output

```
Silhouette Score : 0.8124
```

Graph / Visualization

Dendrogram visualization